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COURSE:DSA0602_DATA HANDLING AND VISUAILIZATION

SET-13: Geographic Data Analysis

Aim

To analyze geographic data using Tableau and R by creating a map chart, scatter plot, and table, and to combine them into an interactive dashboard.

Procedure

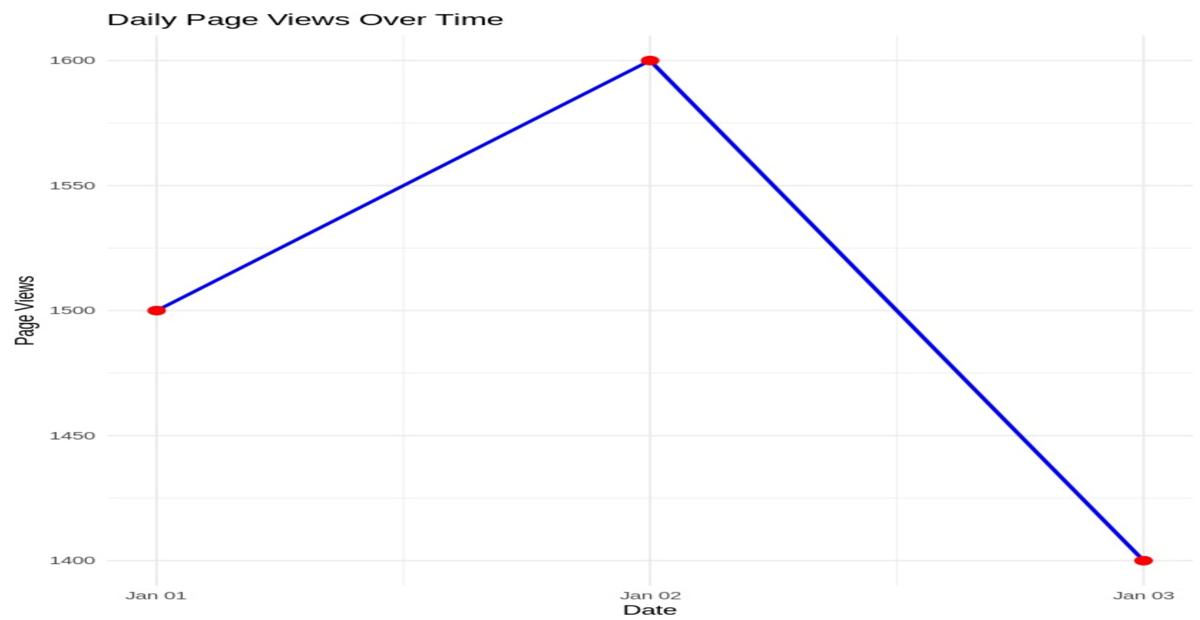
1. Open Tableau and connect to the geographic dataset.
2. Import the dataset containing City, Population, Average Temperature, and Elevation.
3. Create a Map Chart using City as the geographic field.
4. Create a Scatter Plot using Population and Average Temperature.
5. Create a Table displaying all city information.
6. Combine the visualizations into a Dashboard.
7. Analyze the distribution and relationship between variables.

R Code

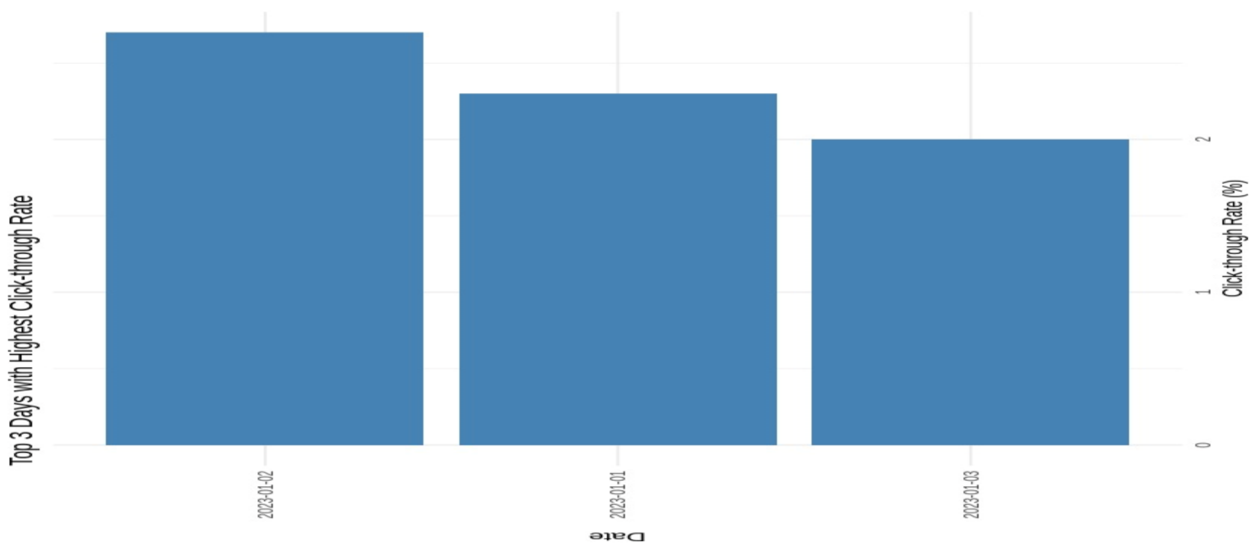
```
library(ggplot2)


city <- c("City A","City B","City C")

population <- c(500000,700000,600000)

temperature <- c(75,68,80)

elevation <- c(1000,800,1200)


data <- data.frame(city,population,temperature,elevation)


print(data)
```

```
ggplot(data,aes(x=population,y=temperature,label=city))+
geom_point(size=4,color="blue")+
geom_text(vjust=-1)+
labs(title="Population vs Temperature",
x="Population",
y="Average Temperature")
```

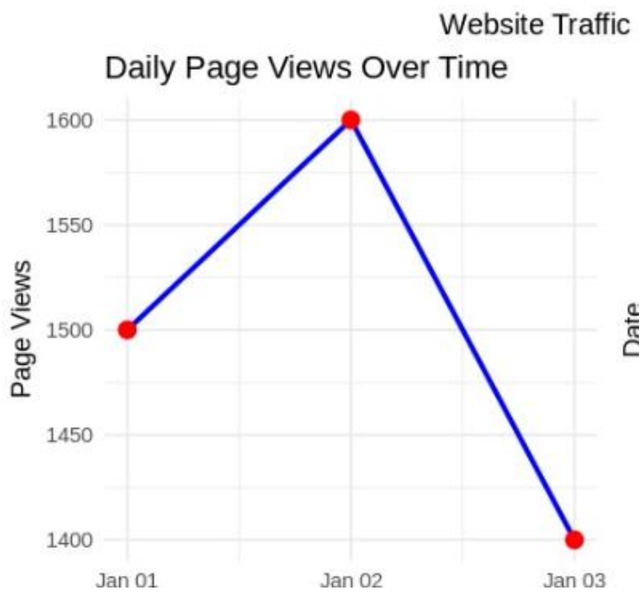
OUTCOME:



OUTCOME 2:



OUTCOME 3:



Result

The geographic dataset was successfully visualized using a map chart, scatter plot, and table. The dashboard enabled interactive exploration of city population, temperature, and elevation.

Experiment 14: Survey Results Analysis

Aim

To analyze survey responses using Tableau and R by creating a stacked bar chart, radar chart, and response table.

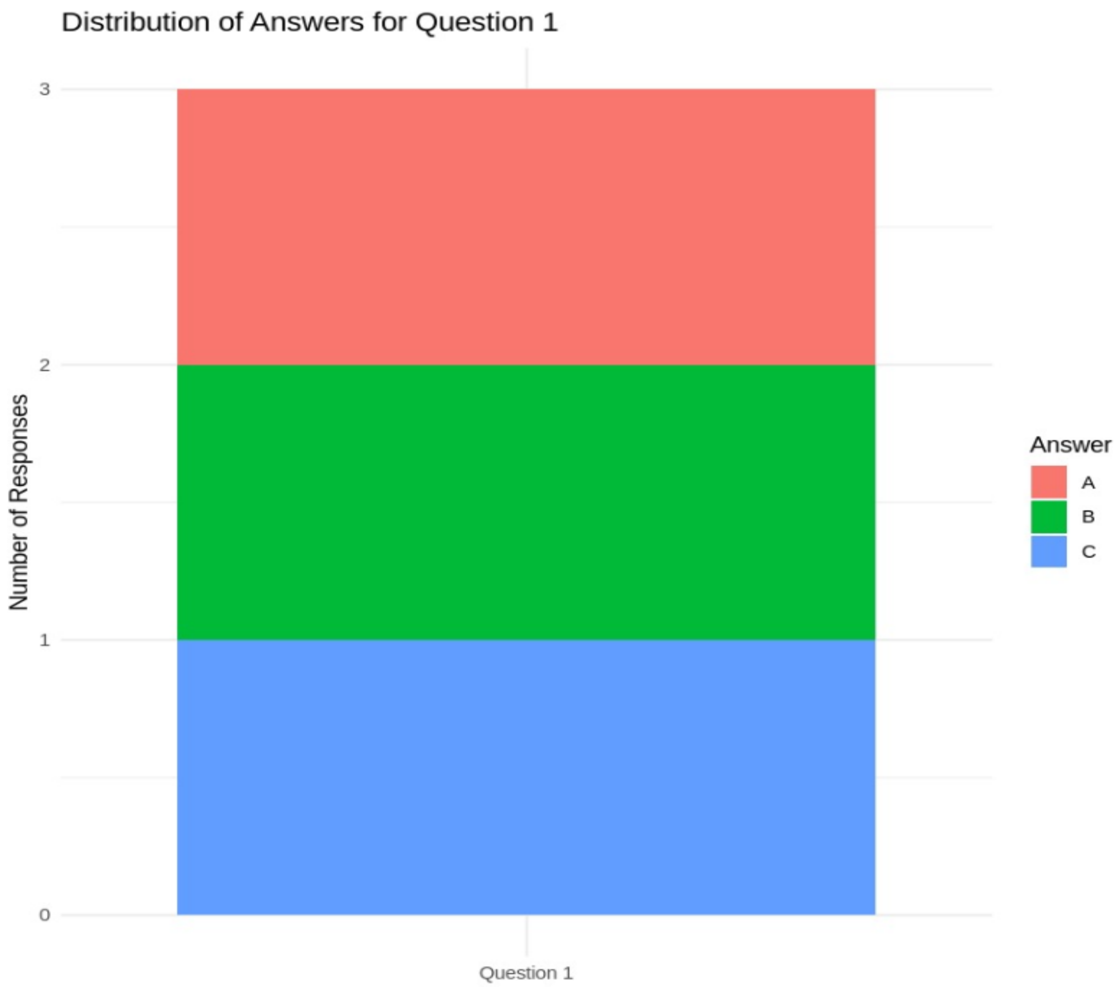
Procedure

1. Open Tableau and connect to the survey dataset.
2. Import the survey response data.
3. Create a Stacked Bar Chart for Question 1.
4. Create a Radar Chart showing responses across all questions.
5. Create a Table containing respondent information.
6. Combine all visualizations into a dashboard.
7. Interpret the survey response patterns.

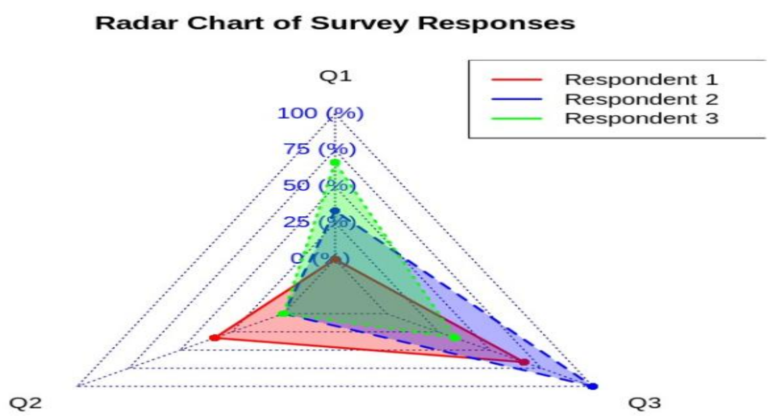
R Code

```
library(ggplot2)


response <- c("A","B","C")

count <- c(1,1,1)


data <- data.frame(response,count)


print(data)


ggplot(data,aes(x=response,y=count,fill=response))+

geom_bar(stat="identity")+

labs(title="Question 1 Responses",

x="Response",

y="Count")
```

OUTCOME1:



OUTCOME 2



Result

The survey responses were successfully analyzed using stacked bar charts, radar charts, and tables. The dashboard provided an interactive summary of respondent answers.

Experiment 15: Product Sales Analysis

Aim

To analyze monthly product sales using Tableau and R by creating grouped bar charts, stacked area charts, and sales tables.

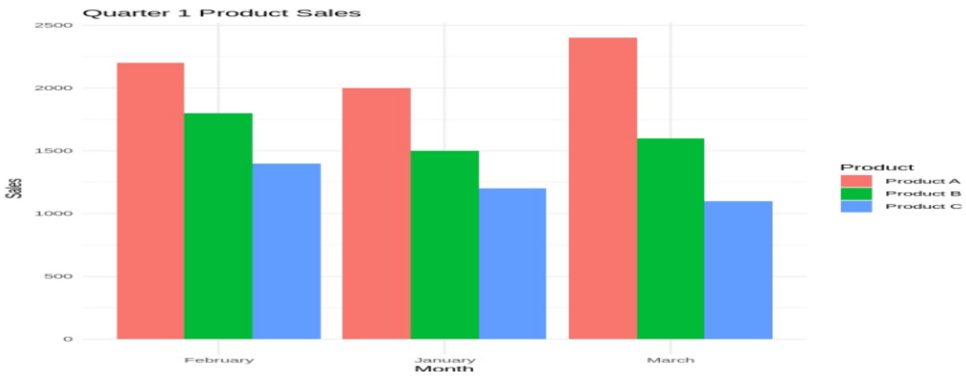
Procedure

1. Connect Tableau to the product sales dataset.
2. Import monthly sales data.
3. Create a Grouped Bar Chart for January, February, and March sales.
4. Create a Stacked Area Chart to show overall sales trends.
5. Create a Sales Table.
6. Combine all worksheets into a dashboard.
7. Analyze product performance.

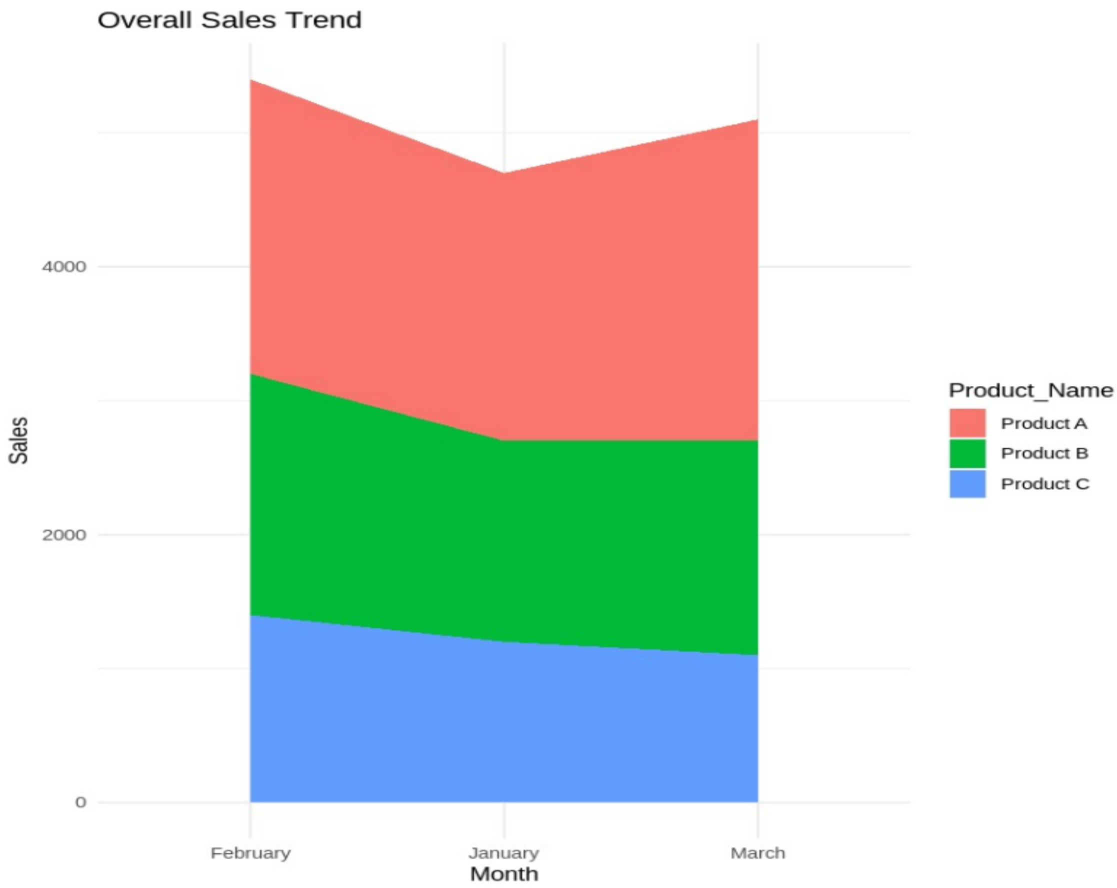
R Code

```
library(ggplot2)


product <- c("Product A","Product B","Product C")
sales <- c(6600,4900,3700)


data <- data.frame(product,sales)


print(data)


ggplot(data,aes(x=product,y=sales,fill=product))+
geom_bar(stat="identity")+
labs(title="Quarterly Product Sales",
x="Product",
y="Sales")
```

OUTCOME 1;



OUTCOME 2:



Result

The grouped bar chart, stacked area chart, and sales table successfully represented monthly product sales. The dashboard enabled easy comparison of product performance.

Experiment 16: Customer Demographics Analysis

Aim

To visualize customer demographic information using Tableau and R by creating bar charts, pie charts, and tables.

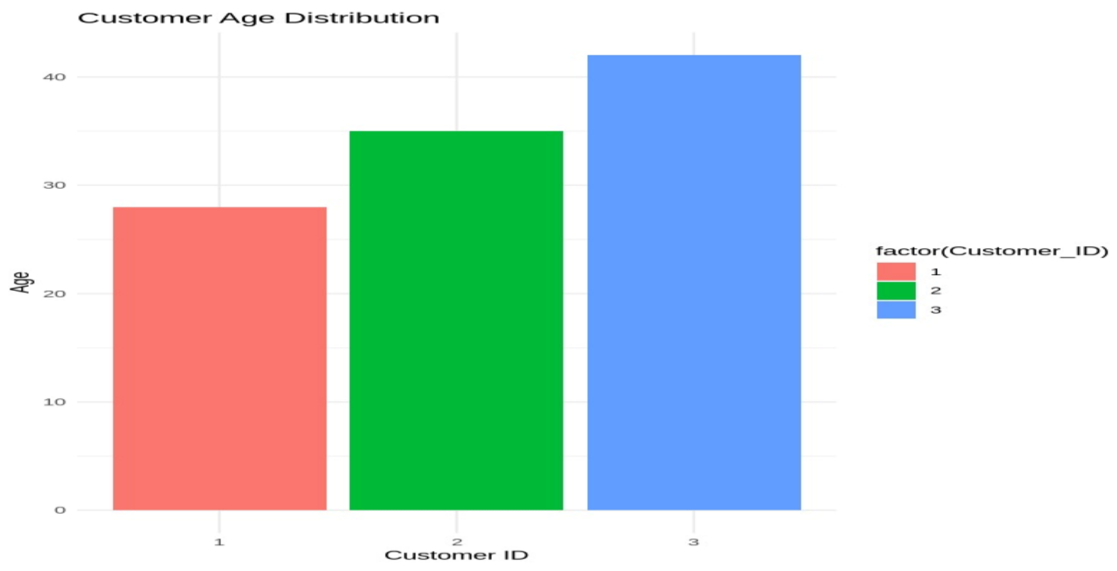
Procedure

1. Connect Tableau to the customer demographics dataset.
2. Import customer information.
3. Create a Bar Chart for customer ages.
4. Create a Pie Chart showing gender distribution.
5. Create a demographic table.
6. Combine all worksheets into a dashboard.
7. Analyze customer demographics.

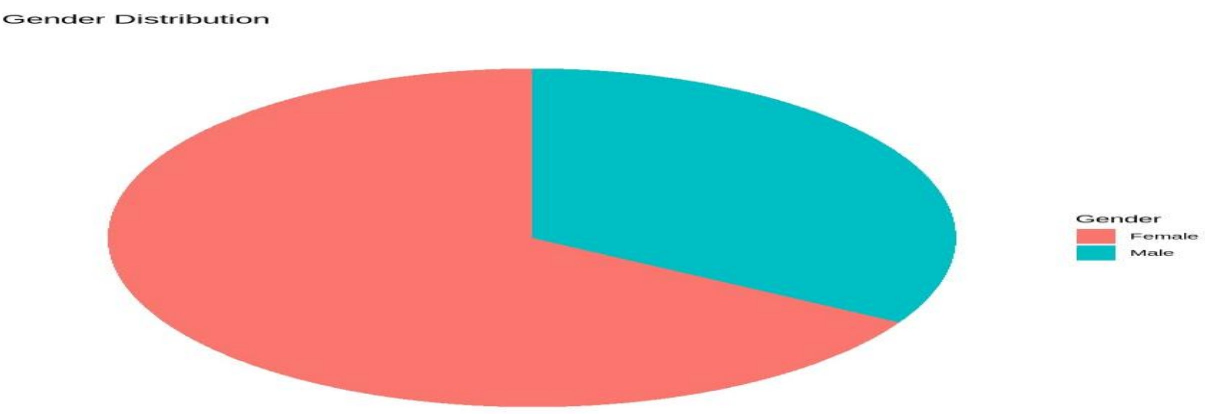
R Code

```
library(ggplot2)

customer <- c("C1","C2","C3")
age <- c(28,35,42)

data <- data.frame(customer,age)

print(data)

ggplot(data,aes(x=customer,y=age,fill=customer))+
geom_bar(stat="identity")+
labs(title="Customer Age Distribution",
x="Customer",
y="Age")
```

OUTCOME 1:



OUTCOME 2



Result

Customer demographic information was successfully visualized using bar charts, pie charts, and tables. The dashboard provided a clear understanding of customer age and gender distribution.

Experiment 17: Employee Performance Analysis

Aim

To analyze employee performance using Tableau and R by creating line charts, bar charts, and tables, and combining them into an interactive dashboard.

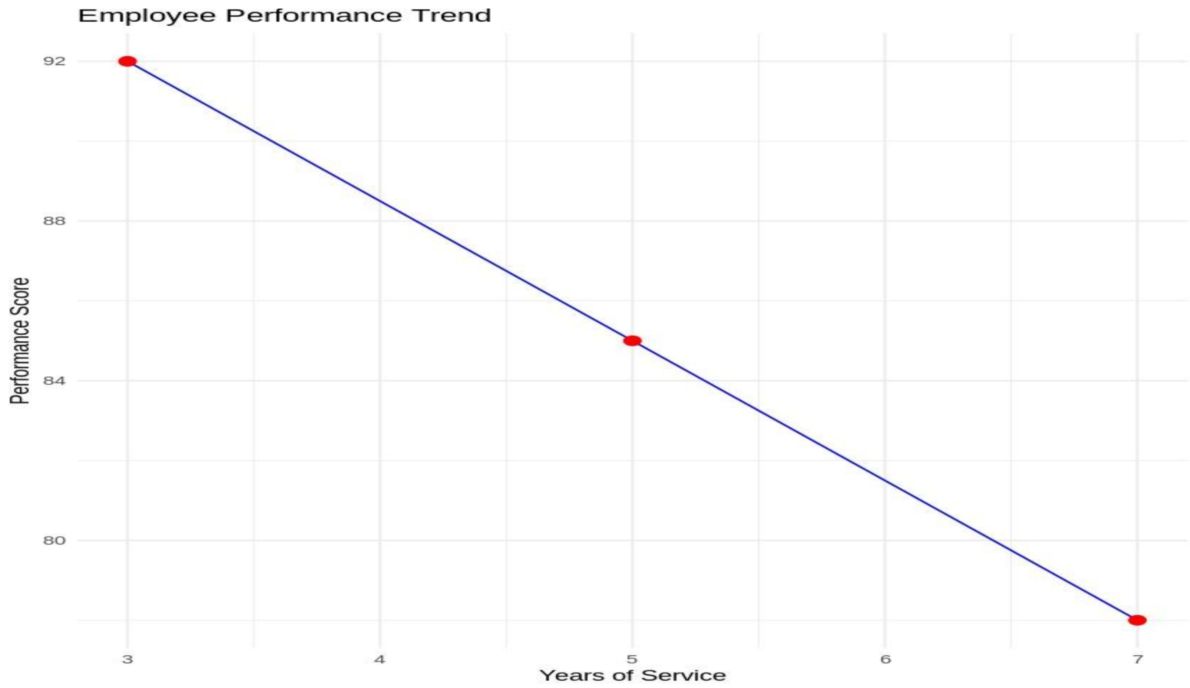
Procedure

1. Connect Tableau to the employee dataset.
2. Import employee performance data.
3. Create a Line Chart for performance scores.
4. Create a Bar Chart showing department distribution.
5. Create a performance table.
6. Combine all worksheets into a dashboard.
7. Analyze employee performance.

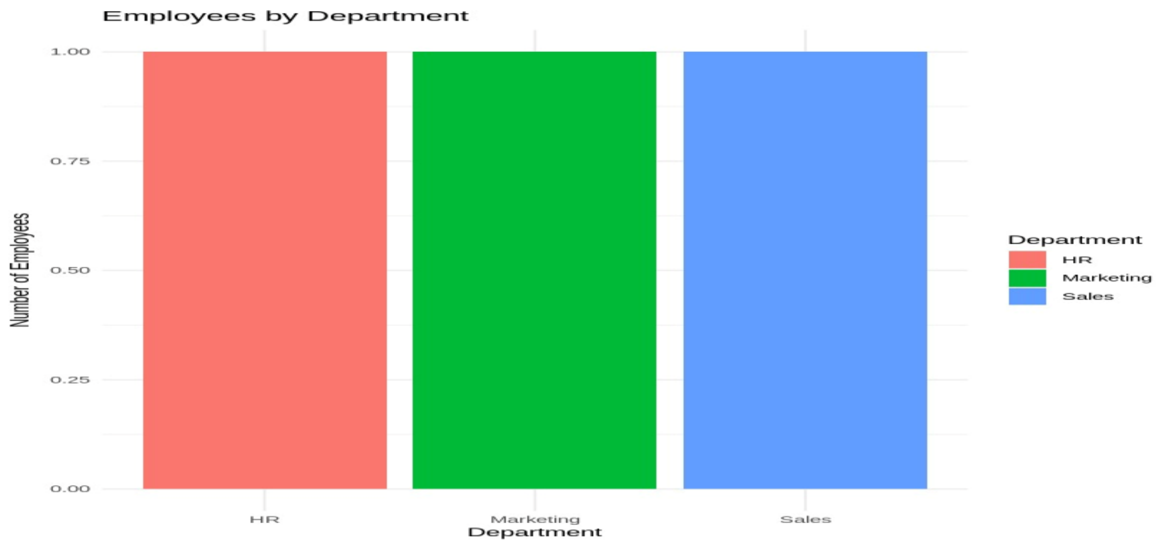
R Code

```
library(ggplot2)


employee <- c("Emp1","Emp2","Emp3")

score <- c(85,92,78)


data <- data.frame(employee,score)


print(data)


ggplot(data,aes(x=employee,y=score,group=1))+
geom_line(color="blue")+
geom_point(size=3)+
labs(title="Employee Performance",
x="Employee",
y="Performance Score")
```

OUTCOME 1:



OUTCOME 2:



Result

Employee performance was successfully analyzed using line charts, bar charts, and tables. The Tableau dashboard provided an interactive view of employee performance and department distribution.

These experiments follow the standard **college record format (Aim → Procedure → R Code → Result)** and are ready to copy into your Word document.